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FC-Trans: Deep learning methods for network intrusion detection in big data environments

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ABSTRACT

With the continuous expansion of Internet traffic, effectively preventing network intrusions in such a vast data environment has become increasingly challenging. Existing intrusion detection systems (IDS) for different network attacks often struggle to identify unknown attacks or respond to them in real-time. In this article, we propose a novel hybrid deep learning model, FC-Trans, designed to enhance network intrusion monitoring. Our approach involves optimizing feature representation using the Feature Tokenizer method, leveraging CNNs to extract meaningful features from the data, and incorporating Transformer's self-attentive mechanism and residual structure to capture long-term feature dependencies and mitigate gradient vanishing. To address the issue of imbalanced sample distribution, we utilize MultiF Loss as the training loss function for the multi-classification task, enabling the model to prioritize difficult-to-classify samples. We compare the performance of our method with other approaches on the UNSW-NB15 dataset, and the experimental results demonstrate significant improvements in both binary and multivariate classification tasks. The results verify the effectiveness of our proposed method.

1. Introduction

With the rapid development of the Internet, the issue of network security has become a growing concern. In the era of big data, network security issues have become more serious and complex, and network intrusion has become a serious threat. Network intrusion can be achieved by means of network attacks and malware, which bring security risks and economic losses to the network, and even endangering national security. Therefore, intrusion detection systems (IDS) have become one of the key technologies for securing networks and are often used to monitor network traffic and issue timely alerts (Liao et al., 2013). Due to the prevalence of network attacks in wide area network (WAN) environments, intrusion detection efforts face significant challenges. Consequently, researchers are compelled to explore new attack detection methods by integrating big data technologies. Intrusion detection is the analysis of packets in the network to discover hidden suspicious behaviors in the data traffic, and alerts and intercepts the intrusions found accordingly. Intrusion detection is essentially a problem of classifying traffic, given the high-dimensional and complex nature of network data traffic. Specifically, it identifies possible attacks and threats by identifying network traffic in a timely manner and determining their specific type.

In recent years, the application of machine learning (ML) in intrusion detection (Sultana et al., 2019; Li et al., 2014; Kuang et al., 2014; Mukherjee and Sharma, 2012) has greatly improved the efficiency and accuracy of detection. However, traditional machine learning algorithms need to rely on complex feature engineering to process the raw data in order to achieve good detection results, and these methods often require experts with relevant knowledge to complete, which is costly, time-consuming, labor-intensive and error-prone. There is also a certain rate of false positives and missed positives. And with the continuous development of deep learning (DL) technology, network intrusion detection (NID) methods based on DL are increasingly used in practical scenarios (Wang et al., 2021; Wu et al., 2020; Kwon et al., 2019). Compared to traditional ML-based network intrusion detection methods, DL-based methods can automatically learn features and patterns from data, reducing manual intervention and false alarm rates, and providing greater robustness and accuracy in detecting new types of intrusions.

There have been many successful applications of DL based network intrusion detection methods. However, there are still some problems with these methods, such as unbalanced samples and insufficient detection of novel intrusions. At the same time, intrusion detection systems

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remain an important research topic as big data technologies continue to evolve and heterogeneous network structures continue to change or update their intruder methods. Therefore, it is important to study how to improve the network intrusion detection capability and robustness of deep learning methods to guarantee network security and protect user privacy, etc.

Based on the above background, combined with the contemporary Transformer architecture that has achieved great success in various fields, we propose a network intrusion detection method based on FC-Trans, aiming to improve the accuracy and robustness of network intrusion detection and contribute to securing the network. The four main contributions of our work are as follows:

(1) We propose an FC-Trans based NID method. By combining the features of convolutional neural networks and self-attentive residual networks, we use convolutional neural networks to extract features with relatively strong local relevance and capture global dependencies between features through a self-attentive mechanism, and map features to higher dimensions through residual structures and feedforward neural networks to improve the representation of data features.

(2) We adopt the Feature Tokenizer method to better combine discrete and continuous features, as an alternative to the traditional one-hot encoding method. This approach significantly reduces the number of feature dimensions while optimizing the feature representation.

(3) We improve the Focal loss function to solve the category imbalance problem in a multi-classification task, so that the model focuses more on samples that are difficult to classify during training and improves the convergence speed of the model.

(4) We evaluate our method on UNSW-NB15 dataset. The results show that the method has good classification performance in both binary and multi-classification tasks and is effective in improving the identification of difficult classification samples.

The remaining sections of this paper are organized as follows. In Section 2, we provide a comprehensive review of related work in the field of intrusion detection. In Section 3, we present a detailed description of the proposed network intrusion detection method. In Section 4, we conduct experiments to analyze and discuss the training process and performance metrics of the proposed method on the UNSW-NB15 dataset. Finally, we conclude this paper in Section 5. The abbreviations used in this article are summarized in Table 1.

Table 1
Summary of abbreviations.

Abbreviations	Notations
IDS	Intrusion Detection System
NID	Network Intrusion Detection
ML	Machine Learning
DL	Deep Learning
SVM	Support Vector Machine
MLP	Multilayer Perceptron
RF	Random Forest
NB	Naive Bayes
GBDT	Gradient Boosting Decision Tree
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GA	Genetic Algorithm
GWO	Grey Wolf Optimization
PSO	Particle Swarm Optimization
FFA	Firefly Optimization Algorithm
ADASYN	Adaptive Synthetic Sampling
FNN	Feed-forward Neural Network

2. Related work

2.1. Machine learning-based intrusion detection

Currently, a large number of NID methods have been applied to distinguish attacks or potential threats from normal traffic. In the early stages of research, traditional machine learning approaches such as Support Vector Machines (SVM) (Gu and Lu, 2021), Random Forest (RF) (Zhang et al., 2008), and k-Means clustering (Muda et al., 2011) were commonly utilized by researchers to construct NID classifiers. However, due to their high false positive rates and susceptibility to overfitting, these methods have consistently been proven to be weak. Aslahi-Shahri et al. (2016) proposed a hybrid model of IDS based on support vector machine and genetic algorithm. The model can reduce the selection of 41 features to 10 features. The proposed method can show excellent true positive value of 0.973 and false positive value of 0.017 on KDD Cup 1999 dataset. Devi and Suganthe (2017) used GWO method to search the feature space to find the optimal subset of features to improve the classification accuracy. The accuracy of the proposed method is tested. The accuracy of the proposed method is compared with the accuracy of several metaheuristic algorithms using NSL-KDD dataset. In addition to this there are some well-known models based on biological meta-heuristic algorithms: PSO (Malik et al., 2015), FFA (Adaniya et al., 2012) etc. are used as feature selection for network intrusion detection algorithms, which improve the performance of NIDS while reducing the number of features. Hu et al. (2008) proposed an intrusion detection method based on the AdaBoost algorithm, a machine learning algorithm that combines multiple weak classifiers into a strong classifier. This approach effectively enhances the accuracy and generalization ability of the classifier. The authors demonstrated the effectiveness of their approach through cross-validation experiments. Liu et al. (2021) proposed an improved NID system that uses a combination of the ADASYN oversampling technique and the LightGBM integrated learning model. In experimental validation on the NSL-KDD, UNSW-NB15 and CICIDS2017 datasets, the results showed that after ADASYN oversampling, the detection rate of a few samples was improved, thus improving the overall accuracy. Furthermore, a variety of GBDT variant models, including XGBoost (Bhattacharya et al., 2020; Devan and Khare, 2020; Kilincer et al., 2022) and CatBoost (Nayak et al., 2022), have been applied in the field of intrusion detection, all of which demonstrate strong performance. However, these tree-based methods overly rely on parameter selection when faced with diverse datasets, warranting further research and exploration in practical applications. Moreover, traditional machine learning methods exhibit a relatively shallow structural complexity. They primarily focus on learning feature importance, feature availability, and dimensionality reduction techniques. While their goal is to uncover optimal correlations among data points that exert the most significant influence on the final outcome, they tend to overlook the crucial inter-feature correlations and neglect to consider the potential impacts of incorporating time steps into predictions. Given the intricate landscape of contemporary heterogeneous networks, these methods struggle to identify unknown attack types and provide real-time solutions. Hence, there is a pressing demand for robust machine learning techniques capable of effectively managing dynamic intrusions as they evolve over time.

2.2. Deep learning-based intrusion detection

With the continuous development of DL, more and more DL-based models achieve excellent performance in NID. Li et al. (2017) converted 41-dimensional features of the NSL-KDD dataset into 8*8 pixel gray-scale maps as input features for a convolutional neural network (CNN) model, and experimented with the learned ResNet50 model. Although it did not give very good results, it also provided a new idea for NID. Yin et al. (2017) proposed a recurrent neural network (RNN)-based IDS. The RNN was applied to time series data processing in intrusion detection and good results were obtained; however, this method could not solve the long-term dependency relationship between features. Yang and Wang (2019) proposed an improved CNN-IDS model that utilizes a parallel 1D-CNN architecture to extract features, their findings suggest that this model achieves superior performance within

Table 2
Related works on network intrusion detection.

Ref.	Methods	Dataset	Classification scenarios
Aslahi-Shahri et al. (2016)	GA-SVM	KDD Cup 1999	Binary
Devi and Suganthe (2017)	GWO-SVM	NSL-KDD	Binary
Hu et al. (2008)	AdaBoost	KDD Cup 1999	Binary
Liu et al. (2021)	ADASYN+LightGBM	NSL-KDD, UNSW-NB15, CICIDS2017	Binary
Li et al. (2017)	ResNet50	NSL-KDD	Binary
Yin et al. (2017)	RNN	NSL-KDD	Binary, Multi-class
Yang and Wang (2019)	ICNN	NSL-KDD	Multi-class
Yao et al. (2021)	CNN-LSTM	KDD Cup 1999, NSL-KDD	Multi-class
Hassan et al. (2020)	CNN-WDLSTM	UNSW-NB15	Binary, Multi-class
Sinha and Manollas (2020)	CNN-BiLSTM	NSL-KDD, UNSW-NB15	Binary, Multi-class
Zhiqiang et al. (2022)	EECA-LSTM	NSL-KDD, CICIDS2017, UNSW-NB15	Binary

shorter time frames. Yao et al. (2021) proposed a cross-layer aggregation CNN-LSTM model. By integrating the features extracted from CNN and LSTM, this model addresses the shortcomings of the serial CNN-LSTM approach, resulting in a comprehensive feature representation encompassing multiple domains. Hassan et al. (2020) employed CNN and weighted discarded long and short-term memory (WDLSTM) networks as feature extractors, using CNN to extract meaningful features from the data, and then WDLSTM to retain long-term dependencies between features, before passing the output to the fully connected layer for classification. The hybrid models combine the advantages of different deep learning techniques in order to improve the accuracy and efficiency of intrusion detection. Sinha and Manollas (Sinha and Manollas, 2020) proposed a deep learning approach that combines CNN and BiLSTM to integrate the learning of spatial and temporal features of the data. The performance of the model was evaluated through cross-validation on the NSL-KDD and UNSW-NB15 datasets. The experiments demonstrated that this model achieves a high detection rate and a low false positive rate. In response to the challenges posed by wireless sensor network environments, Zhiqiang et al. (2022) have introduced an enhanced empirical-based component analysis approach to address feature selection. This feature selection model integrates the strengths of empirical mode decomposition and principal component analysis, effectively retaining a majority of pertinent features. By employing LSTM, the chosen attack nodes were classified. The proposed system was validated on datasets including NSL-KDD, CICIDS 2017 and UNSW-NB15. Comparative analysis unequivocally demonstrates the superiority of the proposed method in terms of performance metrics over existing approaches. Table 2 shows the research results of some scholars in recent years.

However, they still need to perform some complex data preprocessing to achieve data normalization before feeding it into the model. In a current real big data environment, some seemingly simple data operations introduced into the actual environment can frequently lead to an immeasurable computational load. Unlike previous work, we design a new FC-Trans method that quickly fuses different types of features and feeds them directly into the model. The FC-Trans model with a multi-headed self-attentive mechanism can parallelize the computation of each head due to the lack of circular connections, and can better capture the dependencies of different time series in the input sequence without the need to go by time as in the RNN-based model sequence recursively, making training more efficient. FC-Trans is demonstrated to be a feasible NID method by its performance on the UNSW-NB15 dataset.

3. Proposed FC-trans based NID method

In this section, we introduce the NID approach based on FC-Trans (Feature Tokenizer + CNN + Transformer), which is a modification of the Transformer architecture (Vaswani et al., 2017). The framework of proposed model is shown in Fig. 1. This method combines categorical and continuous features from the original dataset using the Feature Tokenizer method. CNN is used to extract locally correlated features and construct multiple spatial feature subsets, with parameter sharing and spatial arrangement. Then, the global feature dependencies are captured by the Transformer's multi-head self-attention mechanism, and the feature relationships are mapped to a higher dimension through a feed-forward neural network (FNN). Finally, the features are classified and outputted through a fully connected layer. Algorithm 1 summarizes the process of this method. The details of each stage in the proposed method are described below.

Algorithm 1 Pseudocode of the Proposed FC-Trans-Based NID Method

Input:
Training dataset D_a
Validation dataset D_b
Testing dataset D_c
Output:
FC-Trans model for intrusion detection
procedure FEATURE TOKENIZER(D_a, D_b, D_c)
    Convert the symbolic features in dataset
    Transform category label to a numeric representation
    $D_e, D_f, D_g \leftarrow$ Integrate the features of D_a, D_b, D_c dataset by Feature Tokenizer
    return D_e, D_f, D_g $\Rightarrow$ new k-dimensional feature space
end procedure
procedure CNN_TRANS(D_e, D_f, D_g)
    while $i \leq n$ **do**
        Load CNN_Trans model
        Input D_e into the CNN_Trans model for training
        if the task is binary classification detection, **then**
            Use binary cross-entropy as loss function
        else
            Use MultiF Loss as loss function
        end if
        Save $MultiFi$ model
    end while
    Save $MultiFn$ model as CNN_Trans model
    Test and save the performance of CNN_Trans model on D_f, D_g
    return CNN_Trans model
end procedure

3.1. Feature tokenizer

Considering that One-hot encoding of categorical features in the data would produce great redundancy, and that even using feature dimensionality reduction would cause the data to lose a great deal of representational information. we introduced the Feature Tokenizer method, The Feature Tokenizer was first applied to the field of tabular data processing and was improved by Gorishniy et al. (2021) to significantly improve the performance of processing tabular data. As shown in Fig. 2, the Feature Tokenizer module combines the input feature x

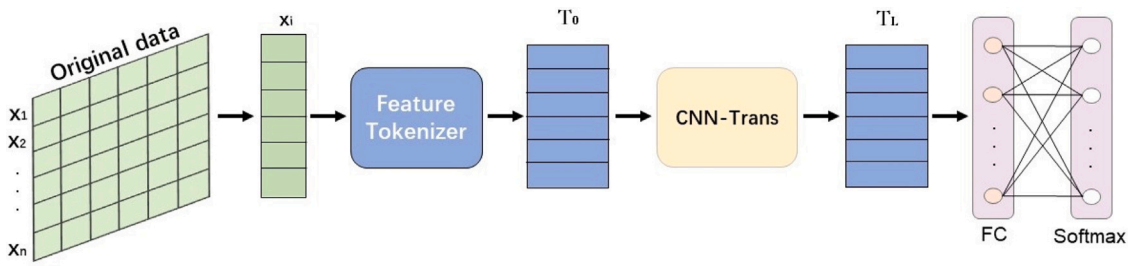

Fig. 1. Proposed framework for FC-Trans-based NID method.

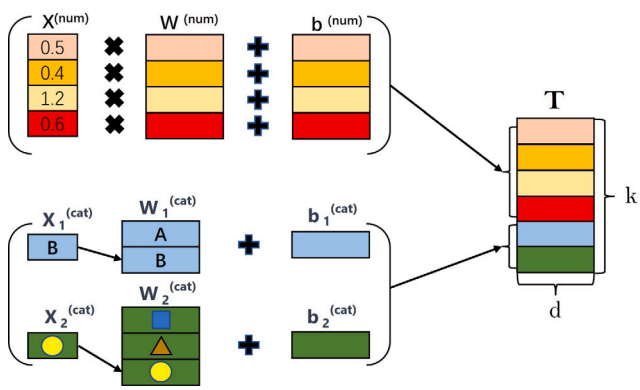

Fig. 2. Feature Tokenizer.

into $T \in \mathbb{R}^{k \times d}$. The fusion calculation of feature x_j is as follows:

$$T_j = b_j + f_j\left(x_j\right) \in \mathbb{R}^d \qquad f_j : X_j \to \mathbb{R}^d \tag{1}$$

where b_j is the jth feature bias, $f_j^{(num)}$ is implemented as the element-wise multiplication with the vector $W_j^{(num)} \in \mathbb{R}^d$ and $f_j^{(cat)}$ is implemented as the lookup table $W_j^{(cat)} \in \mathbb{R}^{S_j \times d}$ for categorical features. Overall:

$$T_j^{(num)} = b_j^{(num)} + x_j^{(num)} \cdot W_j^{(num)} \qquad \in \mathbb{R}^d \tag{2}$$

$$T_j^{(cat)} = b_j^{(cat)} + e_j^T \cdot W_j^{(cat)} \qquad \in \mathbb{R}^d \tag{3}$$

$$T = \text{stack}\left[T_1^{(\text{num})}, \ldots, T_{k^{(\text{num})}}^{(\text{num})}, T_1^{(\text{cat})}, \ldots, T_{k^{(\text{cat})}}^{(\text{cat})}\right] \in \mathbb{R}^{k \times d} \tag{4}$$

3.2. CNN-Trans

The Transformer was first used to solve NLP domain problems, so it has a more complex structure, and its decoder module is an integral part of dealing with NLP problems, but it has no benefit when solving intrusion detection problems, so we made a simple change to the structure of the Transformer, we only kept the encoder module and adapted the structure within the module, as shown in Fig. 3. See Vaswani et al. (2017) for more information on the multiheaded self-attentive mechanism and the computational procedure of the feedforward neural network module. In order to obtain better performance we removed a normalization layer and used the Pre-Norm approach (Wang et al., 2019) instead of the Post-Norm normalization method in the original model. Since the normalization operation in the Pre-Norm is performed before the residual connection, it can improve the normalization of the input signal to a certain level and reduce the subsequent error propagation, thus facilitating the convergence of the model. It can well prevent the gradient explosion or gradient disappearance of the model.

We are inspired by Yang and Wang (2019) that uses a 1D-CNN to compute a spatial attention map of salient and important regions. The aim is to discard unnecessary feature spaces to highlight relevant features and thus learn the relationships between features at the spatial level. Reduces training time and prevents overfitting. The convolution process equation is defined as:

$$y_k = \sigma\left(\sum_{i \in S_k} y_i w_{ik}^c + b_i\right) \tag{5}$$

$$T_0 = H_{concat}\left(G_{\text{flatten}}\left(y_1, y_2, \ldots, y_k\right)\right) \tag{6}$$

where y_k denotes the result of the kth convolutional kernel process, S is the batch block, σ is the activation function, w is the weight and b is the bias. G_{flatten} denotes the flattening process, H_{concat} denotes the cascade operation and T_0 is the extracted multi-spatial feature.

The final token is generated by the CNN and fed into the Transformer structure after the above changes, which is mapped to the high-dimensional space by the L-layer Transformer. The process is as follows:

$$T_e = F_e\left(T_{e-1}\right) \tag{7}$$

This gives T_L by mapping the features to a high dimensional space via CNN-Trans. The final classification is done through the fully connected layer. The process is as follows:

$$\hat{y}_i = \text{softmax}\left(z\right) = \text{softmax}\left(W^T \cdot x + b\right) \tag{8}$$

where x is the input to the fully connected layer, i.e. T_L. W is the weight, b is the bias, and $\hat{y}_i$ is the probability of the Softmax output.

3.3. MultiF loss

Most existing datasets suffer from inherent class imbalance issues, where certain categories are disproportionately represented compared to others. This imbalance in class distribution has consequences such as inadequate training for minority classes, leading to ineffective model training and overfitting. In recent years, in response to this challenge, several notable loss functions have been proposed to address the problem (Lin et al., 2017; Li et al., 2019). Drawing inspiration from the design of Focal Loss, we extended its application to our multi-classification task, setting different weights α_t for each classification sample depending on the amount of data in it. The MultiF Loss is calculated as:

$$MultiF\left(p_t\right) = -\alpha_t\left(1 - p_t\right)^\beta \cdot log\left(p_t\right) \tag{9}$$

where p_t is the estimated probability of the model for class t. β is the degree of loss value decay. The higher the accuracy of the instances, the greater the loss value decay. By incorporating Focal loss into the standard cross-entropy loss, we dynamically adjust the weights of the classes during the training process. This allows the model to pay more attention to challenging classification samples during training. When $\beta = 0$ and $\alpha_t = 1$, the MultiF Loss is equivalent to the standard cross-entropy loss.

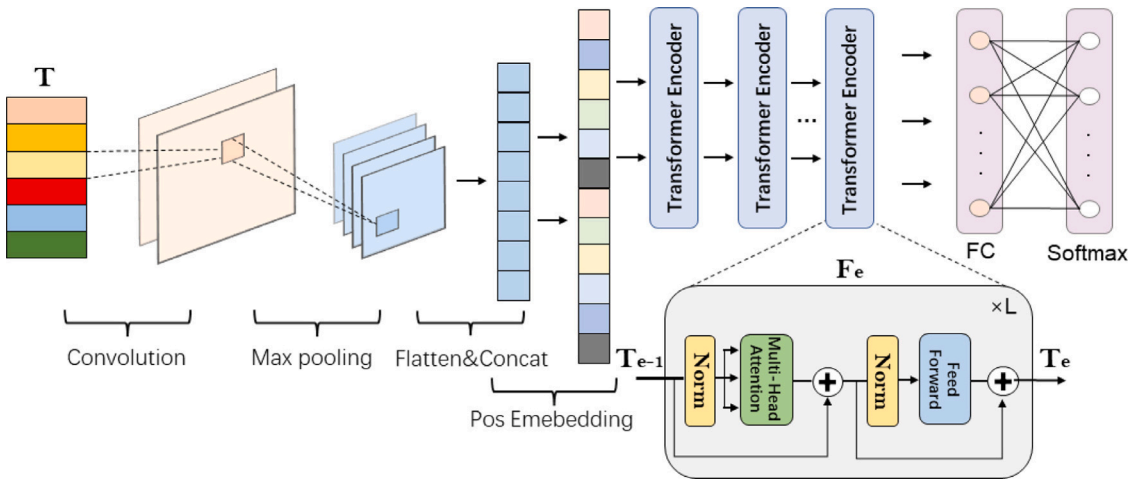

Fig. 3. CNN-Trans Hybrid network architecture.

Table 3
Experimental environment configuration.

Hardware configuration	Software configuration
AMD Ryzen 7 5800H	Ubuntu 20.04
16.0GB RAM	Python 3.8
NVIDIA GeForce RTX 3060	Pytorch 11.3
512G SSD	Numpy 1.24.2

Table 4
UNSW-NB15 Dataset.

Dataset	Category	Samples
UNSW-NB15	Normal	93 000
	Fuzzers	24 246
	Analysis	2677
	Backdoor	2329
	DoS	16 353
	Exploits	44 525
	Generic	58 871
	Reconnaissance	13 987
	Shellcode	1511
	Worms	174
	Total	257 673

4. Experiments, results, and discussions

In this section, we design a three-stage experiment to evaluate the performance of the FC-Trans-based NID method on the UNSW-NB15 dataset. We used MLP, ResNet and CNN-BiLSTM as comparative DL-based models. For a valid and accurate experimental evaluation, we ran three experiments using different random seeds for each model to avoid randomness, and the three runs of average performance as the final result. To evaluate the performance of the proposed FC-Trans model, we conducted a series of experiments divided into three stages. Firstly, we evaluated the performance gains from using the Feature Tokenizer method over other methods. Secondly, we compared the detection performance of various deep learning methods in both binary and multiclass classification tasks. Lastly, we conducted a comprehensive analysis and discussion of the loss and confusion matrix of the FC-Trans model using the standard cross-entropy loss function and MultiF Loss.

4.1. Experimental setup

In order to make the experimental assessment clearer, we present the experimental setting and assessment indicators as follows.

4.1.1. Experimental environment

All experimental models were implemented using Python 3.8, Pytorch framework version 11.3, running on an Ubuntu 20.04 system with NVIDIA GeForce RTX 3060 graphics card, 16gb RAM, 512G SSD. The configuration is shown in Table 3.

4.1.2. Dataset and dataset setup

We first describe the UNSW-NB15 dataset (Moustafa and Slay, 2015) used for the FC-Trans-based NID model. The original network packets for this dataset were created by researchers from the Network Security Research Group at the University of New South Wales, Australia. The dataset consists of over 2 million network flows, which are sequences of packets between source and destination IP addresses. There are nine different types of attacks in the dataset, including DoS, scanning and infiltration. In addition, the dataset includes detailed information about each flow, including protocol type, source and destination port numbers, and the number of packets and bytes in the flow. The dataset was created to facilitate research in the field of network intrusion detection and to provide a benchmark for evaluating the performance of intrusion detection systems. It has been used extensively in a number of studies and competitions, and continues to be a valuable resource for researchers in the field. Table 4 shows the different categories of common attacks in some of the datasets.

Also due to the category imbalance problem in this dataset, different data partitioning methods tend to bias the experimental results. Therefore, we used a uniform partitioning method as a baseline to evaluate the performance of each model in the following experiments. The dataset was partitioned into a training set and a test set according to 8:2. The test set was used for testing only. The training set was oversampled using an oversampling technique for a few classes of Worm classification to balance the training dataset, again using an 8:2 partition into a training and a test set. The test set is used for testing only. The training set is oversampled using an oversampling technique to over-sample a small number of classes of Worm classification to balance the training dataset, again using an 8:2 split to use 20% of the training set as the validation set for tuning and optimization, so that the test set is never used for tuning. The partitioned data are shown in Table 5.

4.1.3. Hyperparameter selection

For all comparison methods, we used the AdamW optimizer (Loshchilov and Hutter, 2017). We use a predefined batch size for all algorithms, we use the optimal loss method for training, training was completed when there was no improvement in accuracy in the validation set for patience+1 epochs. We set the value of patience to 16 for all algorithms. For the proposed FC-Trans method we make

Table 5
Dataset setup.

Dataset	Category	Training	Valid	Testing
UNSW-NB15	Normal	59 520	14 880	18 600
	Fuzzers	15 518	3879	4849
	Analysis	1714	428	535
	Backdoor	1490	373	466
	DoS	10 466	2616	3271
	Exploits	28 496	7124	8905
	Generic	37 677	9420	11 774
	Reco.	8951	2238	2798
	Shellcode	967	242	302
	Worms	59 520	14 880	35
	Total	224 319	56 080	51 535

further optimizations, considering that the computational effort of the attention structure increases quadratically with the number of tokens, we do not complicate the model structure and computational parameters by pursuing its accuracy, and our model parameters are shown in Table 6. This makes its overall performance higher than other proposed models while the computational efficiency of the model is sufficient to be comparable to other current mainstream models.

4.1.4. Evaluation metrics

The parameters used for evaluation and the corresponding calculation methods are as follows: where T_p is a true positive case, T_n is a true negative case, F_p is a false positive case, and F_n is a false negative case. According to the above definitions, $Accuracy$, $DetectionRate$, $Precision$, TPR, FPR, F_1 can be obtained:

$$\text{Accuracy} = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \tag{10}$$

$$\text{Detection Rate} = \frac{T_p}{T_p + F_n} \tag{11}$$

$$\text{Precision} = \frac{T_p}{T_p + F_p} \tag{12}$$

$$\text{False Positive Rate} = \frac{F_p}{F_p + T_n} \tag{13}$$

$$F_1 = \frac{2 \text{ Precision} \cdot \text{Detection Rate}}{(\text{Precision} + \text{Detection Rate})} \tag{14}$$

Of these indicators, $Accuracy$ (ACC) is the ratio of correctly classified samples to the total sample. While this is the most intuitive metric to assess, it is not applicable to situations where the sample distribution is highly unbalanced. $Precision$ is the percentage of samples that the model predicts to be positive that are actually positive, indicating the detection capability of the model. $DetectionRate$ (DR) is defined as the proportion of samples detected as positive to the actual number of positive samples. $FalsePositiveRate$ (FPR) is the percentage of normal records categorized as assaults. The F_1 value is a harmonic mean of $Precision$ and $DetectionRate$, weighted by both (the larger the better) so that model performance can also be determined by F_1, which takes both into account.

4.2. Performance analysis

4.2.1. Evaluation of the feature tokenizer module

In the first phase of experiments, we initially explored the impact of employing 42 features generated through the Feature Tokenizer fusion approach and 196 features generated through the One-hot encoding preprocessing approach on the multi-class accuracy using the CNN-Trans method. The results are presented in Fig. 4. The CNN-Trans method employing the Feature Tokenizer approach showcased superior classification accuracy across all categories compared to the approach employing One-hot encoding preprocessing. This observation suggests

Table 6
Hyperparameter configuration.

Hyper-parameter	Value
cnn_in_channels	42
cnn_kernel_size	2
token	42
head	6
layer	3
learning rate	0.0001
activation function	reglu
decay	10^{-5}
total parameters	57 419

Table 7
Comparison between Feature Tokenizer and various feature optimization methods.

Model	Number of features	$Accuracy$/%	F_1/%	Time/s
CNN-Trans with PSO	25	79.73	76.92	859
CNN-Trans with GWO	19	80.90	78.14	1220
CNN-Trans with FFA	20	80.24	77.35	1117
CNN-Trans with R13	29	81.12	78.33	816
CNN-Trans with One-hot	196	82.82	80.84	7870
FC-Trans	42	**83.01**	**81.21**	1145

that the Feature Tokenizer-based CNN-Trans method highlights more distinct perceptual boundaries across categories, demonstrating its ability to better capture flow features in various scenarios. We also conducted a comprehensive comparison with the optimal feature subsets of several commonly used feature selection methods mentioned in Almomani (2020), along with the method proposed in the original work. As illustrated in Table 7, owing to the significantly reduced feature count employed by the Feature Tokenizer method and the assorted feature selection techniques, a distinct advantage was gained in terms of training time as compared to the usage of the One-hot encoding approach. Worth noting is that several feature selection methods slightly lag behind in various metrics, particularly those utilizing the GWO and FFA feature selection methods. These methods exhibited a slower convergence rate compared to others, underscoring the fact that, for deep learning, the quantity of features is not a matter of fewer being better or more being superior. Instead, emphasis should be placed on the inter-feature dependencies. This firmly establishes the effectiveness of the Feature Tokenizer method in eliminating feature redundancy while substantially reducing feature dimensions and optimizing feature representation. Consequently, the classifier can achieve more effective classification, thereby enhancing the overall performance of the model.

4.2.2. Classification performance of the FC-Trans model

As a misuse detection method for NID, the FC-Trans model exhibits the capability to detect normal and malicious traffic in binary classification tasks. Moreover, in multi-class classification tasks, our model can classify normal traffic and specific attack scenarios. Binary classification, which involves distinguishing normal and malicious traffic, serves as the fundamental task of the NID model. Timely and accurate detection of malicious traffic in a network is essential. For training the binary classification model, we utilize binary cross-entropy as the loss function on the training dataset, while for the multi-class classification model, we employ MultiF Loss as the training dataset's loss function. In this phase of experimentation, we first compare and analyze the performance of the FC-Trans model against other mainstream models in binary classification tasks.

Table 8 presents a detailed overview of the performance of various binary classification models. The experimental results indicate that all four DL models exhibit impressive performance in terms of binary classification accuracy and F1 score, with our proposed model gaining a slight advantage. Of particular note is the False Positive Rate (FPR), which represents the percentage of normal class samples erroneously classified as attack class. In the field of intrusion detection, FPR, along with DR and ACC, collectively form a highly important evaluation

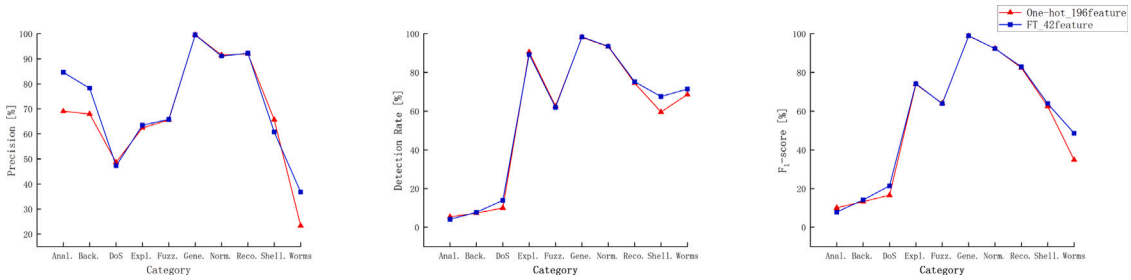

Fig. 4. The precision of the two methods is compared.

Table 8
Comparison of FC-Trans and other deep learning models in binary classification.

Model	$Accuracy/\%$	$FPR/\%$	$F_1/\%$
MLP	92.77	4.58	92.80
ResNet	**94.14**	**2.85**	94.18
CNN-BiLSTM (Sinha and Manollas, 2020)	93.08	5.90	93.58
FC-Trans	**94.27**	**2.99**	**94.28**

metric system. While a model may demonstrate commendable DR and ACC, an elevated FPR suggests challenges in identifying normal samples, thereby impacting its overall effectiveness. Table 8 clearly demonstrates that our proposed model maintains a lower FPR. This aspect underscores the suitability of our model for implementation within intrusion detection systems.

The previous experimental results show that the dl-based approach can achieve good detection performance in both binary classification tasks. However, accurate identifying of different attack types has been a challenge in intrusion detection research due to the uneven distribution of samples in the training dataset, coupled with the similarity of some attacks. In order to more fully evaluate the performance of our proposed FC-Trans model, we compared our proposed CNN-Trans model using the MultiF loss with standard cross-entropy loss and other models in terms of Accuracy, Parameter quantity, $macro - F_1$ and $weighted - F_1$ in multi-class tasks. In the calculation, $weighted - F_1$ is calculated by taking a weighted average based on the sample quantity of each traffic type, while $macro - F_1$ directly calculates the average F_1 score for each traffic type. The evaluation results are provided in Table 9. It is evident that the FC-Trans utilizing the MultiF Loss achieves the best performance across all evaluation metrics. Notably, our model showcases a significant advantage in terms of computational parameter count, considerably fewer than other models. This outcome further underscores the exceptional efficiency of our model design in parameter utilization. In comparison, the FC-Trans model employing the standard cross-entropy loss function also attains commendable classification accuracy results. However, due to its inability to effectively address categories with fewer samples, it results in a relatively lower $macro - F_1$ score. Through the incorporation of the MultiF Loss, our model directs greater attention towards the classification of minority class samples, thereby significantly enhancing the overall $macro - F_1$ performance.

4.3. Discussion

The imbalance in the dataset has a significant impact on the model's training capability. As the training process is influenced by the majority class data, the consequence is that classes with fewer instances receive insufficient training, leading to a decline in the model's generalization ability. This results in ineffective model training and overfitting, which is manifested by the slow convergence of the loss function during the training process. We further analyzed the training loss of the FC-Trans model using the standard cross-entropy loss function and the MultiF Loss. As shown in Fig. 5, it can be observed that as the number of training epochs increases, the loss values gradually decrease and eventually

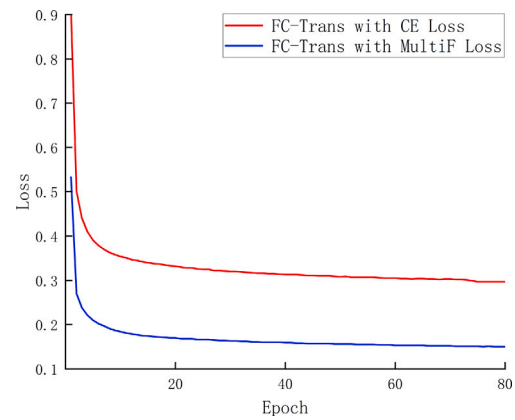

Fig. 5. FC-Trans employs the loss of two loss functions.

converge to near-zero. However, the model trained with the MultiF Loss demonstrates significantly improved loss values during training compared to the model trained with the standard cross-entropy loss. This indicates that employing MultiF Loss in FC-Trans can effectively guide the model to better learn from minority classes, emphasizing the training of challenging samples and resulting in improved performance.

From Fig. 6, it can be observed that the FC-Trans model trained with the cross-entropy loss function predicts many attack categories as "Exploits" attacks on the UNSW-NB15 dataset. The main reason for these misclassifications is that many attacks have similar characteristics, and some attack features often occur together, leading to misjudgment by the model. Another reason is that the classifier lacks attention to difficult samples. In contrast, using the MultiF loss as the training loss function significantly reduced the number of test samples predicted as "Exploits" attacks while being of other categories. The accuracy of predicting "Dos" and "Fuzzers" increased by 10%, and the accuracy of other categories also slightly improved. It is evident that using MultiF Loss as the training loss function enhances the performance of the FC-Trans model in multi-class intrusion detection.

Based on the experimental results mentioned above, our proposed model exhibits high accuracy in detecting various types of attacks, ensuring system security.

5. Conclusion

We propose an FC-Trans network intrusion detection method for big data environments. The Feature Tokenizer method is utilized to optimize the feature representation of data, and meaningful features are extracted using CNN. The Transformer's self-attention mechanism and residual structure are then used to extract the dependency relationships between features and map them to a high-dimensional space for classification using a classifier. For the multi-class classification task, we employ the MultiF Loss to train our model, emphasizing the classification of challenging samples and addressing the issue of

Table 9
Comparison of FC-Trans and other deep learning models in multiple classification.

Model	$Accuracy/\%$	$weighted-F_1/\%$	$macro-F_1/\%$	Parameters
MLP	79.36	76.16	41.89	704 269
ResNet	82.38	**80.01**	**52.14**	7 473 154
CNN-BiLSTM (Sinha and Manollas, 2020)	82.08	76.18	43.74	206 028
FC-Trans with CE	82.51	79.95	48.63	57 419
FC-Trans with MultiF	**83.01**	**81.21**	**56.79**	**57419**

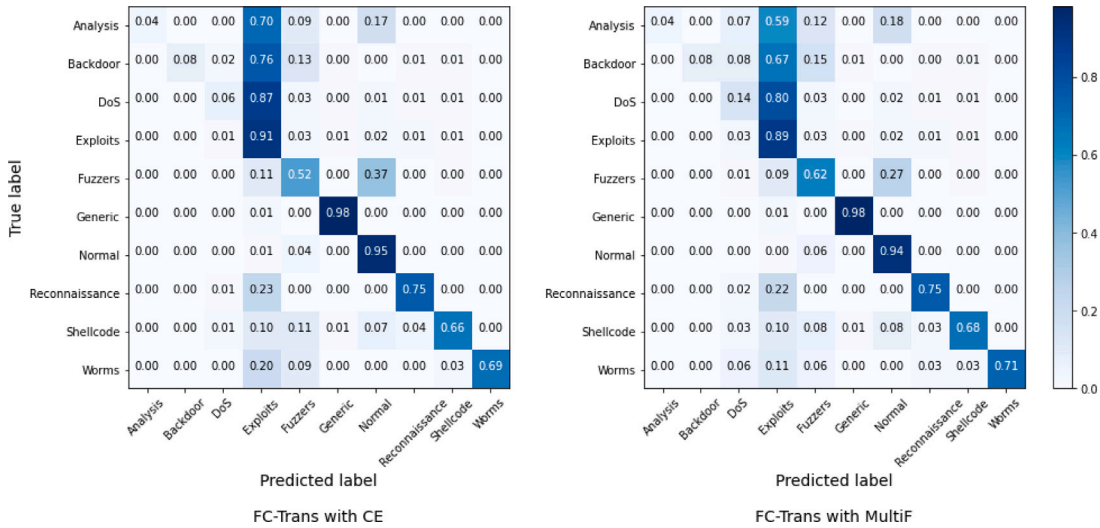

Fig. 6. Confusion matrices for standard cross-entropy Loss and MultiF Loss for FC-Trans in multiclass classification.

imbalanced sample distribution. Experimental results demonstrate that our model exhibits excellent intrusion detection performance in both binary and multi-class classification tasks. Therefore, our approach proves to be a viable solution. In our future research, we will prioritize the practical deployment of intrusion detection methods, with a specific focus on addressing intrusion detection applications in IoT devices.

CRediT authorship contribution statement

Yuedi Zhu: Conceptualization, Methodology, Validation, Software, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Yong Wang:** Funding acquisition, Supervision, Resources, Formal analysis, Writing – review & editing. **Lin Zhou:** Supervision, Project administration, Review & editing. **Yuan Xia:** Visualization, Software, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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